

Enabling Multi-modal Conversational Interface for Clinical Imaging

Kailas Dayanandan
Indian Institute of Technology
Delhi, India
kailasd@gmail.com

Brejesh Lall
Indian Institute of Technology
Delhi, India
brejesh@ee.iitd.ac.in

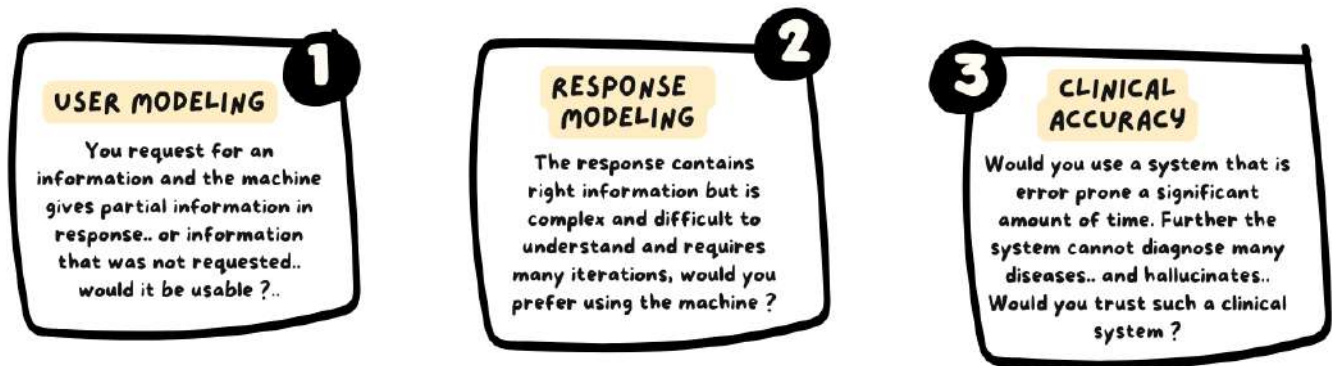


Figure 1: Focus areas that can improve the usability of conversational interface in clinical setting in hypothetical AI radiologist. Our first research question focus on understanding clinician's requirements better (RQ.1), second research question focus on easier access of information by providing optimum information and removing redundant information (RQ.2) and third research question focus on system designs that enhance usability and their impact on clinical accuracy (RQ.3).

ABSTRACT

Human-computer interaction research has to play a vital role in increasing the adoption of deep learning models in clinical settings, as their adoption is low despite models surpassing/matching the clinician's performance on many medical imaging tasks. Conversational AI has been successful as an interface for general information; however, there is a research gap for multi-modal conversational interface design for safety-critical clinical imaging systems. Our research points to the important role of multi-modal chat in improving usability and explainability through textual and visual explanations. Our main contributions include design principles for conversational interfaces in clinical imaging systems, the importance of multi-modal responses, and an understanding of the usefulness of mimicking clinician/radiologist interactions to improve usability. We show that diagnosis descriptions and visual responses improve the multi-modal conversational interface. The multi-modal conversational interface can help improve the adoption of deep learning systems in clinical settings, improving clinicians' efficiency and patient outcomes.

CCS CONCEPTS

• Human-centered computing → Natural language interfaces;

KEYWORDS

Conversational AI, Generative AI, Healthcare, Chest X-Ray

ACM Reference Format:

Kailas Dayanandan and Brejesh Lall. 2024. Enabling Multi-modal Conversational Interface for Clinical Imaging. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/3613905.3650805>

1 INTRODUCTION

Deep learning models have matched the performances of trained clinicians on many medical tasks across various specialties, including cardiology [24], ophthalmology [59], neurology [65], radiology [3] and pulmonology [62]. The AI systems and clinicians working together showed better performance and lower diagnosis time [13, 43, 52], which emphasizes the importance of improving the adoption of AI-based systems. Despite good accuracy, they find limited use in clinical settings, and significant reasons for low adoption include lack of explainability [21, 23, 53] and poor human-computer interaction [38]. The conversational interface can also serve as an explainability tool that increases transparency in model decision-making, provides better interpretation of model behavior [39, 64], and contribute to efforts in improving explainability [16, 70, 79]. The lack of user modeling led computer aided diagnosis systems to

focus on diseases that the radiologist could easily identify rather than those that are hard for the radiologist [38], and poor response modeling turned the systems to be a burden, slowing them with several false predictions [38]. Our study on user/response modeling and clinical accuracy (fig.??) align with effort and performance expectancy in the unified theory of acceptance and use of technology model (UTAUT model) for a general information system [12, 18, 69] and are essential for understanding design principles for conversational interface for clinical imaging systems [29]. The studies on diagnostic systems provide insights about user behavior; for example, empirical studies have established a preference for diagnosis explanations [9, 36] and mimicking the expert language [47]. However, they lack depth as they have not explored further to understand 'why' users prefer them and the merits of such behavior. There is a research gap in the design of conversational interfaces that provide explainable diagnoses with the right amount of information in the apt format to enhance their usability in busy clinical settings. Our goal is to identify design principles to improve the usability of multi-modal conversational interfaces for clinical imaging systems, which brings to our research questions

RQ. 1 : What are the common queries/intents in clinician interaction with a clinical imaging system?

RQ. 2 : How do multi-modal conversational interfaces improve the usability of clinical imaging systems?

RQ. 3 : How does system design impact usability in a conversational interface?

In our study, we use thematic analysis to understand the patterns of information the clinician requests. The analysis of the clinician's queries resulted in the identification of eight patterns of questions. The multi-modal responses are then formed by combining diagnosis descriptions used by radiologists (textual component) and matching the intent/information from the question with visualizations that best communicate the requested information (visual component). Using relational content analysis, we analyze diagnosis descriptions and their contextual importance to disease and observe the nuances in clinician-radiologist communication. We use user trials to verify this diagnosis description as the preferred form of textual content by comparing them with short text similar to diagnostic systems and an expanded version generated using a large language model. The participants found the visual component in the response to be complementary, along with diagnosis descriptions, though some parts could be dispensable.

Our contributions are (a) We identify the information requirement of clinicians from clinical imaging systems (b) Our empirical studies show that visual components in conversation improve usability and the natural language explanations used by radiologists based on disease are optimal (c) Our study shows that clinical performance of end to end natural language systems are inferior to existing diagnostic systems but can generate appropriate supporting information.

2 RELATED WORK

Human-computer interaction research studied text-based conversational agents [71] in general settings on many tasks whose behavior differs from clinical imaging systems. The research on conversational agents mainly focuses on support roles post-disease diagnosis (e.g. support for patients with traumatic brain injury [28], dementia[60], cardiovascular diseases [50], alcohol use disorder [58], palliative care [61], improving mental health [7, 19, 20, 32, 42, 45, 51]), where the most probable information can be the most appropriate information in the response. In the case of conversational interfaces in diagnostic systems, the information is patient-specific about how the image differs from a normal condition, which is not generic. The recent success of ChatGPT as a reliable conversational interface and the performance of large language models on medical tasks [63] inspired many systems in medical domain, which includes conversational agents [77, 81, 81] and diagnostic systems [44, 68, 78, 87]. The human interaction research on conversational agents concerns empathy [32], persona [57], and related parameters, which may not be necessary for diagnostic systems that focus on communicating clinically correct information to busy clinicians [5] and require specific research for clinical imaging systems. Multi-modal conversational systems in the general domain focus only on retrieving existing images [17, 46, 56, 75, 80], whereas safety-critical systems require generating images that explain decision-making [76]. To the best of our knowledge, the design of conversational interfaces for safety-critical clinical systems that communicate with clinicians is absent in human-computer interaction research.

HCI research in diagnostic systems focuses on non-conversational interfaces but provides insights into clinician's needs. The work on diagnostic systems looked into textual communication of positive or negative diagnoses using rule-based methods (e.g., assertiveness, personalization based on experience, confidence scores) to improve usability, diagnosis time, and diagnostic accuracy [9]. The information can be presented better visually [10], especially for explaining decision-making [1, 76]; hence, visual components can be helpful in conversational systems. A response containing relevant and optimal information can reduce time/iterations to gather the information, improving usability in busy clinical settings. The clinician may be interested in a disease's severity and other factors based on the disease, requiring prioritization of information. The studies focus only on evidence from co-occurrence [36], confidence [9], and location [76]. Textual and visual responses are studied separately; however, they can be complementary or redundant together [37] in conversational interfaces. Research on design principles or usability of conversational interfaces for safety-critical clinical systems is currently absent. Medical Report Generation and Visual question answering for medical imaging are technically similar to conversational interfaces as they generate textual information from images [7, 30, 67, 73, 85], but are not optimized for factual correctness [49, 85, 86], and suffer from hallucination [31, 82, 88], which is unacceptable in clinical settings. Models that match clinician's performance operate directly on images to diagnose diseases and have evolved to handle peculiarities of disease distribution [11, 15, 25, 25–27, 72, 84]. The conversational systems should match the accuracy of existing diagnostic systems, as usability and explainability improvements should not be at the cost of clinical accuracy.

3 METHODS

In order to address the first research question, we find the common types of questions that the clinician is likely to query. The common types of questions are estimated using a question-answer dataset (section 3.1.1) containing questions by clinicians on clinical images (RQ.1). This helps generate an appropriate visual response based on query type and also helps improve the coverage of the system responses. We obtain the best visual component in the response by matching the information in the visual responses with information sought in the clinician's query (RQ.2)

The information is relevant if it is time-consuming for the clinician to understand, whereas it is redundant if the information is generic and the same for all the patients. We use the details about the affected area and the location for diseases in annotated datasets (section 3.1.3) to analyze the relevance of information in disease diagnosis explanations. This analysis of relevance is significant as our study concerns clinical systems for busy clinicians. We use the medical reports written by radiologists for clinicians in the medical report dataset (section 3.1.2) to verify the presence of useful details and avoidance of redundant information (RQ.2).

Diagnostic accuracy is essential in clinical systems, and we look into the diagnostic capabilities of systems that help generate better explanations (RQ.3). The observations from user and response modeling, including optimal textual representation and inclusion of visual components, are studied qualitatively using semi-structured interviews in the final part of our study (section 3.2).

3.1 Datasets

3.1.1 Question Answer Dataset : VQA-RAD is a publicly available visual question-answering dataset containing 2248 questions asked by clinicians about radiology images (head axial single-slice CTs or MRIs, chest x-rays, abdominal axial CTs) and their reference answers [41]. Fifteen volunteer clinical fellows from the National Institutes of Health (NIH) and students who had completed core medical school rotations participated in the study. The participants provided "free-form" questions that mimicked them asking a colleague or another physician for 20 images and rephrased questions by other participants for 20 images.

3.1.2 Medical Report Dataset : MIMIC-CXR contains 3,77,110 chest x-rays and associated semi-structured free-text radiology reports from 227,835 imaging studies conducted on 65,379 patients at the Beth Israel Deaconess Medical Center Emergency Department in Boston [33] and has many derivatives with additional information [6, 40, 48, 66]. MIMIC-CXR is a benchmark dataset for medical report generation tasks for chest X-rays and is accessible free of cost after credentialing.

3.1.3 Disease Annotated Dataset : VinDr-CXR is a publicly available dataset with 18,000 postero-anterior (PA) chest X-rays collected from Hospital 108 and the Hanoi Medical University Hospital [55]. 17 radiologists annotated the affected region for 14 critical radiographic findings with each x-ray independently analyzed by at least three radiologists.

3.2 User Study

We studied the clinical support system by conducting semi-structured interviews with fourteen clinicians. During the interview, the clinicians were presented with the prototype with a multi-modal conversational interface and encouraged to discuss the reason for their decisions, and the interviews lasted 45 minutes to 1.5 hours.

3.2.1 Prototype. We implement a web application using a medical report generation model [67], which generates different sentences from a chest x-ray. We form the text part of the multi-modal response by selecting the sentence most similar to the question. The similarity between the sentences and the questions are found using BERT score [83], and we use a threshold of 0.9 to ensure correct sentence selection. The use of sentences similar to medical report-style diagnosis explanation helps replicate the optimal response. We use template-based matching to generate the query type (Appendix Table. 3) and obtain the disease using text matching, which helps form the correct visual response (Table. 1, Fig. 2). We use ground truth bounding boxes annotated by the clinicians in our study. An example of the multi-modal response generated by the prototype is in fig 3. The visual responses that are constant are pre-determined and stored (e.g., information about disease in fig.2d, contrastive explanations in fig.2c). The settings in the web application help toggle between textual and multi-modal responses during the interview.

3.2.2 Study Design. The interviews started with participants writing five questions, which they would likely ask a clinical imaging system. We then show the participants sample questions from the identified types based on codes in the thematic analysis and possible textual responses, including template-based and diagnosis explanations, to gather feedback on the utility of supporting information. We study the optimal form of the textual components by comparing the template-based response in diagnostic systems, explanations as in medical reports, and additional information incorporated using large language models. The interviews also focus on the effectiveness of multi-modal conversations, as the visual component can be complementary or redundant, along with diagnosis descriptions. The participants compared prototypes of text-based and proposed multi-modal conversational interfaces for precise comparisons. We finally interviewed the participants on the possible use of conversational interfaces in existing diagnostic systems to improve the usability and usefulness of visual components in medical reports.

3.2.3 Participants. The participants were recruited either from the institute hospital (eight clinicians) or were referred by the lab members to which the authors belong (six clinicians). Among the participants, nine had an advanced qualification, and seven were senior clinicians with over ten years of experience (Refer to Table 2 for more details).

3.3 Data Analysis

3.3.1 User modelling : We did a thematic analysis of the questions in the VQA-RAD dataset to identify common pattern of questions. We use inductive coding to identify the intent-based codes from the data. These codes help understand the type of questions the system can receive in the conversational interface. The explainability-related questions from the XAI question bank [47] that are relevant for clinical imaging are also included in the analysis.

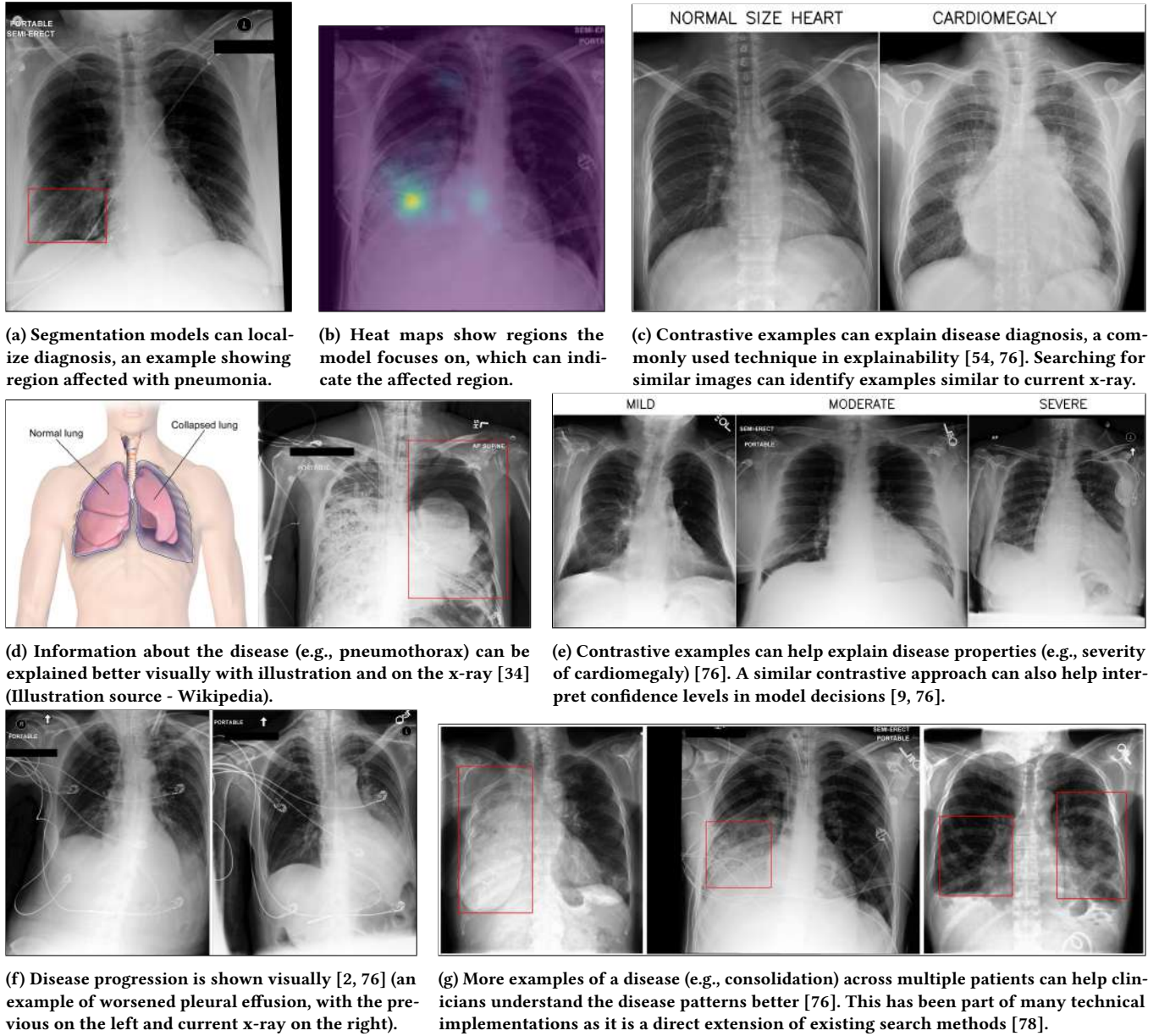
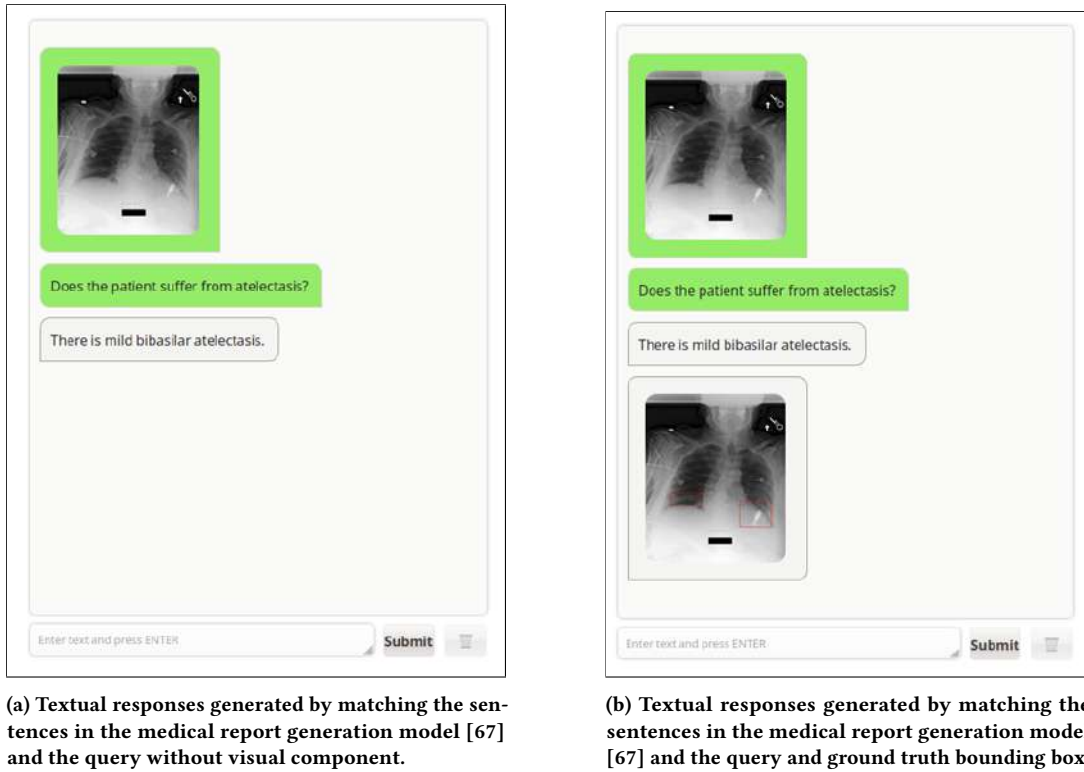


Figure 2: Visual responses can present information better and require the right choice of visual response based on the question. Visual responses like segmentation bounding boxes or heat maps can convey localization information. Recent diffusion-based generative AI allows severity images to be generated from x-rays under study [14, 74].

3.3.2 Response modelling : We did a relational content analysis of medical reports to understand the nuances in communication between radiologists and clinicians. We analyze the relationship between words used along with diseases to understand patterns and their usefulness. We further conducted a content analysis of reports generated by deep learning systems to understand patterns and diagnostic accuracy. We also verify the applicability of the observations from medical reports and our method of generating visual responses by content analysis of interview transcripts.

4 FINDINGS

In this section, we start by understanding clinicians' requirements and then discuss the importance of the optimal form of textual presentation and the usefulness of visual response for easier comprehension. We then discuss the formation of multi-modal responses, followed by a discussion on clinical accuracy and precision for the clinical imaging systems.



(a) Textual responses generated by matching the sentences in the medical report generation model [67] and the query without visual component.

(b) Textual responses generated by matching the sentences in the medical report generation model [67] and the query and ground truth bounding box.

Figure 3: A screenshot from the prototype showing the difference between textual and multi-modal responses. The textual response with the diagnosis description is more usable as the details are present in a first response rather than letting the user query again for these details.

4.1 Clinician Conversations

The main insights from the analysis to identify the type of information requested by a clinician and the usability of multi-modal responses are

(1) Thematic analysis and inductive coding of the VQA-RAD dataset resulted in seven types of questions a clinical imaging system can receive from clinicians (Table 1). In our interviews, we collected possible questions from the clinicians, and they matched the types of questions identified from the VQA-RAD dataset. The participant's questions reflected their approach to interpreting X-rays by starting with the organ studied in the X-ray, the view of the X-ray, followed by the analysis of different regions, and this information was rare in medical reports. The reports contain only the findings from these analyses, and one participant's comment illustrated their thinking process "... which view is this whether it is AP or PA.. we know that.. we don't know whether we will be asking it.. being a doctor.. (P7)". The questions also showed the VQA-RAD dataset's effectiveness in understanding clinician's questions (Table 1). We added questions about comparison with previous x-rays, which were not supported in the VQA-RAD dataset. We also added probing questions from explainability question bank and The clinicians found questions related to diagnosis, location, and severity most helpful "... diagnosis and location is the most important.. (P2)".

They expressed satisfaction with the coverage of questions identified in our analysis. *"these questions are useful.. these are fine..(P2)"*. Many participants did not find the questions on system capabilities from the explainability question bank useful, indicating that explainability should be more about diagnosis than the system. *".. probing questions are not important.. (P5)"*, which contradicted with existing studies where the user showed interest in understanding system capabilities, limitations, and idiosyncrasies [4, 8]. Another finding is the usefulness of region-based queries, which clinicians use to refer to different chest regions for any missed diagnosis or the presence of any abnormalities. *"..I can ask many leading questions if my diagnosis does not match with the shown diagnosis.. (P4)"*.

(2) In our study, we observed a preference for multi-modal responses containing visual components for both textual components (diagnosis description and template-based responses). The reasons for the preference for visual components include usefulness and interpretability. *".. they can just pinpoint the diagnosis.. so that it is faster and quicker.. especially if you have so many patients.. (P2)"*. Another participant observed that *".. yea, it will be helpful.. the whole purpose is to make the work faster.. and more efficient.. if it is pointing out that area.. rather than the eyes going and searching it will save.. even 2-3 seconds is also a lot.. when you are seeing.. (P8)"*. As clinicians are more likely to miss findings due to perceptual error (80% can be identified retrospectively [22]), location can be

helpful information. The clinician's preference for multi-modal conversations shows the visual component to be complementary to textual content and not redundant. The participants also preferred diagnosis descriptions from the report for *".. being specific and not too detailed .. (P8)"* compared to extended explanations generated from large language models, *".. as I have studied all this in mbbs course.. I do not need these all these details.. (P9)"*. This aligns with existing research that indicates improved usability of systems that capture the mental model of experts [76]. However, one senior clinician who referred us to a specialist before agreeing to take part in our study *".. you should check with Dr. ___.. she has done internal medicine..(P5)"*, preferred a detailed answer *"..for chest physician.. he does not need all this.. but for general physician like me who is not expert in chest x-ray.. I might need this also.. (P5)"*.

We modeled the user's information requirements to identify eight patterns of information that the conversation interface should support (RQ.1). All the clinicians preferred multi-modal conversational systems as their preferred response method, and many clinicians mentioned 'faster diagnosis' and 'easy to use' as the reasons for their choice (RQ.2).

4.2 Multi-modal Responses

Clinicians and radiologists are busy in clinical settings; hence, medical reports should contain the right amount and type of information and avoid irrelevant and redundant information. Our relational content analysis helped identify patterns in supporting information for different diseases from the medical reports, which can help generate better multi-modal responses (RQ.2).

(1) In our content analysis of medical reports with positive diagnoses of nodules, we observe that 81.5% specified the location and omitted in only 18.4% of reports. 7% recommended further investigation using CT, and 6% had multiple nodules. The location heat-map for nodules (fig. 4a) shows that they occur at different regions in the lungs, and the affected region is small in size (fig.4b), making the location information necessary for the clinician to identify the nodule easily. In contrast, the reports with a positive diagnosis for cardiomegaly omitted the location details, as the heart is easily identifiable, making location details redundant. It shows the importance of location information in textual explanation to identify affected regions in diseases like nodules, but it is redundant in cardiomegaly. All the participants agreed about the location being redundant for cardiomegaly in the visual response. *".. the location is not important for cardiomegaly .. (P4)"*. Some participants were fine with showing it for the sake of consistency *".. it can be shown for cardiomegaly also .. (P2)"*, while few others recommended contrastive explanations or textual indication for as the diagnosis method was not same *".. text indication about not using the ratio is better.. (P4)"*. The clinicians use anatomical regions to refer to locations in the queries (Region Query in Table.1), and radiologists use them to indicate affected regions in medical reports.

(2) In our content analysis of medical reports with positive diagnoses of cardiomegaly, we observe that the severity of cardiomegaly is mentioned in 80% of the reports, indicating its importance for the disease. This shows that supporting information varies with disease, and many properties apart from location information are present for specific diseases. The participants found severity to be

helpful in the response. *".. severity is important .. (P4)"*, which could be the reason for better usability with diagnosis descriptions. The participants also found severity and location information useful for disease atelectasis in the prototype during the interview (Appendix fig.3).

(3) The participants preferred the selection of the visual responses by matching the question types to the information contained in the visual response (Table 4). The participants preferred the most appropriate visual component *".. show only the correct one.. you don't need to other images (P4).. "* and also mentioned the selection to be important *".. the visual system is very important if accurate .. (P6)"*, which shows the significance of selecting apt visual components. Some of the participants prioritized the bounding box as the first response for diagnosis-related questions *".. if you know the location is.. you can know around what it is trying to say .. (P7)"*. This also aligns with existing studies about errors made by radiologists' being perceptual errors *".. location details are sufficient.. remaining I can find out..(P9)"*. The participants, however, mentioned different visualizations as useful when explicitly queried *".. all are useful.. (P3)"*. The participants were asked about the least helpful visual response; the comparison with normal images *".. diseased and normal comparison image is least useful..(P2)"* and search *".. examples using search is the least useful .. (P7)"* were selected by most participants.

(4) All the clinicians preferred visualizations in the medical report *".. this is good.. very helpful .. (P5)"* and *".. it would be helpful.. (P7)"*. This finding is significant as a system that can help radiologists generate better reports for clinicians can drive the adoption of AI-based systems. The participants found the visual component also improved usability in template-based responses *".. multi-modal is always the better option .. (P7)"*. The clinicians preferred diagnosis descriptions with visual components over other combinations.

In summary, the participants found the visual components helpful in conversations and emphasized the importance of showing the appropriate visual response (RQ.2). The clinicians preferred short sentences similar to diagnosis details in medical reports as they contained diagnosis details like severity and location specific to analyzed X-ray and omitted general information that clinicians were aware of (RQ.2). The participants pointed out some improvements in the prototype *".. the images are small.. when you click on it.. it should pop out.. or make it bigger.. (P4)"* and *".. the date of the previous x-ray is also required .. (P6)"*.

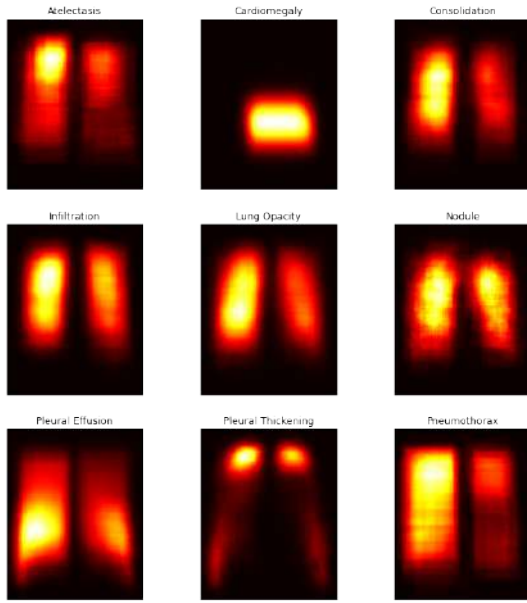
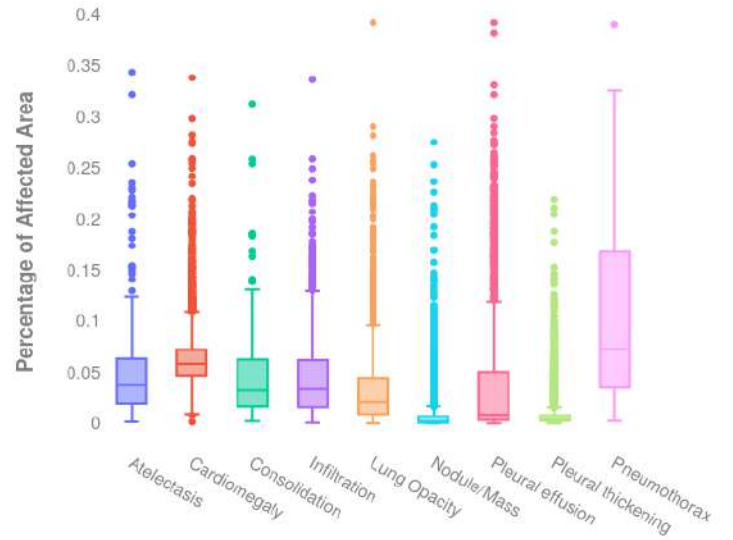
4.3 Accuracy-Usability Trade off

Natural language systems that directly generate text from images have become prominent in recent years. In this section, we study the generation of textual components with disease descriptions using models trained on medical reports and compare them with diagnostic systems that only identify the presence of disease.

(1) Our content analysis of system-generated reports contained details about the severity and omitted location details for cardiomegaly in all the generated reports. This eliminates the need to understand and encode the patterns for different diseases, as these systems can reproduce patterns from reports. These systems are significantly poor in diagnosing rare diseases compared to existing diagnostic systems, implying that the usability improvements came at the cost of accuracy, which is essential in clinical systems. *".. focus*

Table 1: Major types of questions

Question Type	Examples	Visual Explanation
Diagnosis	'Is there a fracture?' [41]	Heatmaps, Segmentation
Location	'Where is the lesion located?' [41]	Heatmaps, Segmentation
Diagnosis and Location	'Is there evidence of a pneumothorax' [41]	Heatmaps, Segmentation
Disease/Organ Attributes	'Are the nodules well circumscribed?' [41]	Heatmaps, Segmentation, Contrastive Explanations
Region Query	'What is in the left apex?' [41]	Heatmaps, Segmentation
Probing	'How do you determine cardiomegaly?' [41]	Search Results, Patient comparison
General	'What is a pleural effusion' [41]	General
Disease Progression	'How has the XX progressed?' [41]	Contrastive Explanations

**(a) Location heat-map distribution showing the regions where different diseases are most likely to occur for VinDr-CXR Dataset [35].****(b) Distribution of the ratio of bounding box of affected region to overall image size.****Figure 4: Plots showing properties of common diseases in VinDr-CXR [55].**

on disease diagnosis.. not anything else .. (P6)". A minor error in the expanded diagnosis generated using a large language model was pointed out by a participant "... bilateral means it affects both lungs equally' is not correct.. one can be more than the other.. (P3)", indicating unreliability for generating additional information. However, processing reports to generate positive diagnoses without adding additional information was error-free.

(2) As false alarms were a main hindrance in existing computer aided diagnosis systems, we asked participants about the number of false alarms tolerable for correct diagnosis. All the participants chose one false alarm per 4 or 5 images "... one false alarm per five is OK ...(P4).. and "...I think 20% false alarm is OK (P3)..

(3) Many participants preferred to prioritize the disease that requires immediate attention compared to observations in existing

studies about prioritizing diagnoses that are hard to find for radiologists "... more fatal diseases should be focused.. else patient will suffer more .. (P6)" and "... you should be highly accurate on life threatening conditions like pneumothorax.. (P4)". However, some participants had views that aligned with existing studies [22, 38] about focusing on difficult to identify diseases "...sometimes if the naked eye cannot see.. if you can do that.. that is what precision you want from technology.. right.. what the naked eye cannot see.. (P8)" or those commonly missed out by radiologists "... if system is finding something.. and we did not find it.. it is more beneficial.. (P9)". The clinicians were also concerned about the cost savings in more accurate diagnosis using AI systems "... the x-ray should find things that is difficult for radiologists like malignant or benign nodules.. if this can be found out

the patients can avoid the costly CT scan .. (P6)". These comments also indicate directions for future research.

We modeled the response for the system to be usable for clinicians and helpful in clinical settings. The clinicians found the system with less than one false alarm for four responses more usable and stressed the importance of diagnostic accuracy (RQ.3). The users preferred the systems to accurately diagnose diseases requiring immediate attention and help improve their accuracy by finding difficult-to-identify diseases (RQ.3).

5 DESIGN IMPLICATIONS

(1) Conversational interface should correctly identify the clinician's needs and provide intuitive and optimal responses. Clinicians prefer the right amount and type of information, and conversational systems should avoid information that clinicians already know. The study found that the existing research on mimicking the expert to increase usability also applies to conversational interfaces. The diagnosis descriptions from medical reports include information relevant to disease diagnosis and avoid redundant information, and this can help form better textual responses in multi-modal chat for imaging systems used in busy clinical settings. Our study also observed that additional relevant supporting information about diagnosis is helpful for the clinician, and its inclusion can result in a more intuitive interface that better captures the clinician's requirements.

(2) Visual components improve conversational interface and medical reports. Clinicians found the visual component complementary despite the detailed information in diagnosis descriptions. Though the clinicians found all types of visual explanations helpful, they preferred only the most appropriate visualization in the response. These responses were obtained by matching the information in the query with the information in the response. Our study also found that visual details can be helpful even in medical reports (fig.5), and the visualizations may be avoided currently due to time constraints or legacy practices. The participants also found heat maps helpful in identifying the affected region (fig.4b), compared to an earlier study where only classification criteria were necessary, and location information was irrelevant [37]. The participants' preferences among different possible visual responses confirmed our process of matching information in visual and textual components with the user intent. Integrating visual responses in the conversational interface can help address usability and explainability concerns in computer-aided diagnostic systems.

(3) Diagnosis explainability is more important over explainability of system capabilities. The clinicians in our study corroborated the findings from our technical analysis that the diagnosis description contains important attributes that vary based on diseases. Our study also observes that report-generation systems can reproduce these nuances in diagnosis descriptions. Clinicians consider diagnosis essential and other related queries like system capabilities as nice to have. As the information is directly accessible without an access point on the screen in a conversational interface, there is an opportunity to include these details and improve the coverage of information available to the clinician.

(4) Conversational system should be accurate with fatal diseases and radiologist's blind spots. The participants had a low

tolerance for loss of accuracy for better usability, underlining the fact that usability of the conversational interface should not come at the cost of poor diagnostic performance. The clinician's preference for an AI system depends on their job priorities, with clinicians mainly preferring systems that focus on fatal diseases, whereas existing studies showed radiologists' preference for systems that identify diseases they are likely to miss out. The clinicians found the systems usable when the false alarm is less than one in four or five predictions, which can be considered in system design to improve adoption and clinician's efficiency. The multi-modal interface also helps with easier comprehension of findings, facilitating acceptance among clinicians and leading to better patient outcomes.

6 CONCLUSION

In this paper, we explored different aspects to improve the usability of a multi-modal conversational interface. The analysis of communication between radiologists and clinicians showed the presence of many nuances that helped convey important information about disease diagnosis and discard redundant information. The analysis of the questions helps to form better multi-modal responses by co-relating the intent with visualizations that contain matching information. The insights can also help improve explainability and usability in a conversational interface, enhancing adoption in clinical settings and improving patient care through faster response times and improved diagnosis. The multi-modal interface also helps with easier comprehension of findings, facilitating acceptance among clinicians and leading to better patient outcomes. Our work has broader implications, including generating conversations to form training data for end-to-end deep learning systems as an alternative to the expensive procedure of engaging clinicians. The approaches in our paper can be helpful for other downstream interfaces and applications (e.g., voice inputs) and other imaging modalities like CT scans, MRI, and mammography.

ACKNOWLEDGMENTS

The authors thank Dr.Hareesh of IIT Delhi Hospital for helping arrange the interviews. The authors also acknowledge the code contributions from the VinDr-CXR competition website for generating plots and for report generation from [67], which we use in the prototype web application.

7 APPENDIX

7.1 Medical Report

Medical report is generated by adding visual information and bounding boxes to indicate the affected region and is used in clinician interviews. The current report is retained in the upper right corner. The participants in the user study are shown in Table 2.

7.2 Generation of Multi-modal Responses

Our study focuses on systems used by clinicians in busy hospital settings. The pattern of questions a conversational interface receives in clinical settings is shown in Table.3. Identification of visual components useful in a multi-modal conversation is an essential step, and fig.2 shows visualizations used in our study, and Table.4 shows information present in different visual responses.

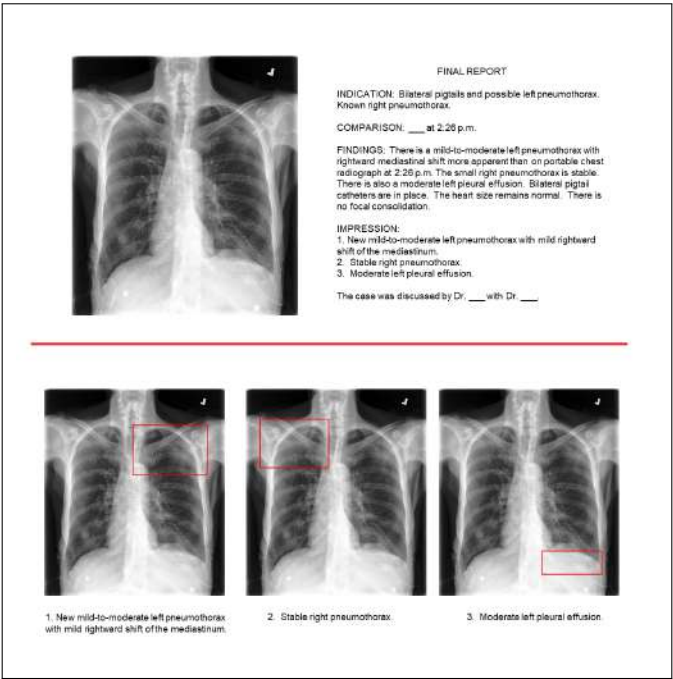


Figure 5: Existing report with X-Ray snapshot and bounding boxes showing affected area.

Table 2: Participant Details

Participant	Specialization	Qualification	Clinical Experience
P1	Cardiac Surgeon	MS, MCh	10+ years
P2	General Surgeon	MS	24 years
P3	General Physician	MBBS	35 years
P4	Anesthesiologist	DNB	10+ years
P5	Emergency Physician	DEM	18+ years
P6	Emergency Physician	DEM, DGM	15+ years
P7	Psychiatrist	MD, DNB	4+ years
P8	General Physician	MBBS	4 years
P9	General Physician	MBBS	3+ years
P10	General Physician	MBBS	3+ years
P11	Oncologist	MS	3+ years
P12	Urologist	MS, MCh	10+ years
P13	General Physician	MBBS	5+ years
P14	General Physician	MD	4 years

The system has to choose the best visual response depending on the information requested in the query and the information in the visual response. The optimal responses are obtained for different queries by combining the related text from medical reports and visual components by matching the information requested and the information in the visualization (Table 1).

The textual component with additional information is obtained using a large language model and relevant sentence from the medical report [67]. The prompt to add information to the disease diagnosis explanations and the output generated by the model’s output

is shown below. The prompt is run on Llama 2, a 7B parameters model fine-tuned for chat responses from Meta (Llama-2-7b-chat).

<>You are a radiologist answering the doctor.
Do not acknowledge my request in any way and
don’t use ‘based’ or ‘sure’ or ‘thank you’ in the
response and go straight to the answer. Don’t
add points you don’t know or add any more
diseases. Avoid introductions. <>[INST]There is

Table 3: Major types of questions

Question Type	Examples
Diagnosis	'Is there a fracture?' [41], 'Why did the system not predict XX' [47]
Location	'Where are the nodules located?' [41]
Diagnosis and Location	'Is there evidence of a pneumothorax?' [41]
Disease/Organ Attributes	'Are the nodules well circumscribed?' [41]
Disease Progression	'How has the XX progressed?' [47]
Region Query	'Can you see all of the ribs on the left side?' [41]
Probing	'Is this an MRI?' [41], 'How accurate/precise/reliable are your predictions?' [47], 'What are the limitations of the system?' [47]
General	'About how often do you see bilateral Wilms tumor?' [41] 'What indicates a pleural effusion?' [41]

Table 4: Types of visual responses and their relation to question types

Visualization	Information	Question Types
Heat maps	Approximate location of diagnosis	Diagnosis, Location, Region Query
Segmentation	Precise location of diagnosis	Diagnosis, Location, Region Query
Counterfactual Explanations	Images with and without a specific disease	Diagnosis, Probing, Disease attributes, General
Decision Interpretations	Images of different severity / confidence levels	Disease severity, Probing, General
Progression comparison	Prior images of the patient	Disease Progression
Patient comparison	Images with similar diseases	Diagnosis, Probing, General
Disease details	Image explaining the disease	General

mild bibasilar Atelectasis. [/INST]. describe only about the disease in X-Ray in detail?

Mild bilateral atelectasis on an X-ray suggests that some of the lung tissue has collapsed, resulting in reduced lung volume. The term "bilateral" means it affects both lungs equally. In this case, there is mild collapse of the lung tissue, indicating partial loss of function. Atelectasis can be caused by various factors such as pneumonia, bronchitis, chronic obstructive pulmonary disease (COPD), or other respiratory conditions that reduce airflow into the lungs. It may also occur due to chest injuries or tumors. The appearance of atelectasis on an X-ray depends on its severity; however, the presence of atelectasis usually results in decreased lung capacity, which can lead to shortness of breath with exertion or at rest if severe enough.

diagnose rare diseases. We chose the tail classes comprising the rarer disease in the MIMIC-CXR dataset for our analysis, and the results are shown in Table 5.

Table 5: Recall for rare diseases in evaluated on MIMIC-CXR for a recent report generation model [67].

Disease	Distribution	Occur	Detect
Pneumomediastinum	0.3	4	0
Fibrosis	0.44	33	0
Emphysema	0.85	64	0
Subcutaneous Emphysema	0.93	17	0
Lesion	0.96	17	0
Tortuous Aorta	1.32	12	0
Hernia	1.53	18	1
Calcification of the Aorta	1.65	1	0
Mass	2.09	41	0
Nodules	2.89	67	0

7.3 Performance on Rare Diseases

The generated reports of patients diagnosed with rare diseases are analyzed to check whether the report generation systems [67] can

REFERENCES

- [1] Agathe Balayn, Natasa Rikalo, Christoph Lofi, Jie Yang, and Alessandro Bozzon. 2022. How can Explainability Methods be Used to Support Bug Identification in Computer Vision Models?. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [2] Shruthi Bannur, Stephanie Hyland, Qianchu Liu, Fernando Perez-Garcia, Maximilian Ilse, Daniel C Castro, Benedikt Boecking, Harshita Sharma, Kenza Bouzid,

- Anja Thieme, et al. 2023. Learning to exploit temporal structure for biomedical vision-language processing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 15016–15027.
- [3] Soumen Basu, Mayank Gupta, Pratyaksha Rana, Pankaj Gupta, and Chetan Arora. 2022. Surpassing the human accuracy: detecting gallbladder cancer from USG images with curriculum learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 20886–20896.
- [4] Emma Beede, Elizabeth Baylor, Fred Hersch, Anna Iurchenko, Lauren Wilcox, Paisan Ruamviboonsuk, and Laura M Vardoulakis. 2020. A human-centered evaluation of a deep learning system deployed in clinics for the detection of diabetic retinopathy. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–12.
- [5] Edward I Bluth, Donald P Frush, M Elizabeth Oates, Seth A Rosenthal, and Claire Bender. 2022. A recommendation for addressing the physician workforce crisis contributing to burnout in radiology and radiation oncology. *Journal of the American College of Radiology* 19, 9 (2022), 1072–1073.
- [6] Benedikt Boecking, Naoto Usuyama, Shruthi Bannur, Daniel C Castro, Anton Schwaighofer, Stephanie Hyland, Maria Wetscherek, Tristan Naumann, Aditya Nori, Javier Alvarez-Valle, et al. 2022. Making the most of text semantics to improve biomedical vision–language processing. In *Computer Vision–ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XXXVI*. Springer, 1–21.
- [7] Eleanor R Burgess, Ivana Jankovic, Melissa Austin, Nancy Cai, Adela Kapuścińska, Suzanne Currie, J Marc Overhage, Erika S Poole, and Jofish Kaye. 2023. Healthcare AI Treatment Decision Support: Design Principles to Enhance Clinician Adoption and Trust. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [8] Carrie J Cai, Samantha Winter, David Steiner, Lauren Wilcox, and Michael Terry. 2019. "Hello AI": uncovering the onboarding needs of medical practitioners for human-AI collaborative decision-making. *Proceedings of the ACM on Human-computer Interaction* 3, CSCW (2019), 1–24.
- [9] Francisco Maria Calisto, João Fernandes, Margarida Morais, Carlos Santiago, João Maria Abrantes, Nuno Nunes, and Jacinto C Nascimento. 2023. Assertiveness-based Agent Communication for a Personalized Medicine on Medical Imaging Diagnosis. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–20.
- [10] Francisco M Calisto, Alfredo Ferreira, Jacinto C Nascimento, and Daniel Gonçalves. 2017. Towards touch-based medical image diagnosis annotation. In *Proceedings of the 2017 ACM International Conference on Interactive Surfaces and Spaces*. 390–395.
- [11] Francisco Maria Calisto, Nuno Nunes, and Jacinto C Nascimento. 2020. BreastScreening: on the use of multi-modality in medical imaging diagnosis. In *Proceedings of the international conference on advanced visual interfaces*. 1–5.
- [12] Francisco Maria Calisto, Nuno Nunes, and Jacinto C Nascimento. 2022. Modeling adoption of intelligent agents in medical imaging. *International Journal of Human-Computer Studies* 168 (2022), 102922.
- [13] Francisco Maria Calisto, Carlos Santiago, Nuno Nunes, and Jacinto C Nascimento. 2022. BreastScreening-AI: Evaluating medical intelligent agents for human-AI interactions. *Artificial Intelligence in Medicine* 127 (2022), 102285.
- [14] Pierre Chambon, Christian Bluethgen, Jean-Benoit Delbrouck, Rogier Van der Sluijs, Małgorzata Polacin, Juan Manuel Zambrano Chaves, Tanishq Mathew Abraham, Shivanshu Purohit, Curtis P Langlotz, and Akshay Chaudhari. 2022. RoentGen: vision-language foundation model for chest x-ray generation. *arXiv preprint arXiv:2211.12737* (2022).
- [15] Bingzhi Chen, Zheng Zhang, Yingjian Li, Guangming Lu, and David Zhang. 2021. Multi-label chest X-ray image classification via semantic similarity graph embedding. *IEEE Transactions on Circuits and Systems for Video Technology* 32, 4 (2021), 2455–2468.
- [16] Haomin Chen, Catalina Gomez, Chien-Ming Huang, and Mathias Unberath. 2022. Explainable medical imaging AI needs human-centered design: guidelines and evidence from a systematic review. *NPJ digital medicine* 5, 1 (2022), 156.
- [17] Xiaolin Chen, Xuemeng Song, Liqiang Jing, Shuo Li, Linmei Hu, and Liqiang Nie. 2023. Multimodal dialog systems with dual knowledge-enhanced generative pretrained language model. *ACM Transactions on Information Systems* 42, 2 (2023), 1–25.
- [18] Fred D Davis, Richard P Bagozzi, and Paul R Warshaw. 1989. User acceptance of computer technology: A comparison of two theoretical models. *Management science* 35, 8 (1989), 982–1003.
- [19] Paula Maia de Souza, Isabella da Costa Pires, Vivian Genaro Motti, Helena Medeiros Caseli, Jair Barbosa Neto, Larissa C Martini, and Vânia Paula de Almeida Neris. 2022. Design recommendations for chatbots to support people with depression. In *Proceedings of the 21st Brazilian Symposium on Human Factors in Computing Systems*. 1–11.
- [20] Chadha Degachi, Myrthe Lotte Tielman, and Mohammed Al Owayyed. 2023. Trust and Perceived Control in Burnout Support Chatbots. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–10.
- [21] Tribikram Dhar, Nilanjan Dey, Surekha Borra, and R Simon Sherratt. 2023. Challenges of Deep Learning in Medical Image Analysis—Improving Explainability and Trust. *IEEE Transactions on Technology and Society* 4, 1 (2023), 68–75.
- [22] Warren B Gefter, Benjamin A Post, and Hiroto Hatabu. 2022. Special Features Commonly Missed Findings on Chest Radiographs: Causes and Consequences. *Chest* (2022).
- [23] Philipp Hacker, Ralf Krestel, Stefan Grundmann, and Felix Naumann. 2020. Explainable AI under contract and tort law: legal incentives and technical challenges. *Artificial Intelligence and Law* 28 (2020), 415–439.
- [24] Awni Y Hannun, Pranav Rajpurkar, Masoumeh Haghpanahi, Geoffrey H Tison, Codie Bourn, Mintu P Turakhia, and Andrew Y Ng. 2019. Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. *Nature medicine* 25, 1 (2019), 65–69.
- [25] Md Inzamam Ul Haque, Abhishek K Dubey, Ioana Danciu, Amy C Justice, Olga S Ovchinnikova, and Jacob D Hinkle. 2023. Effect of image resolution on automated classification of chest X-rays. *Journal of Medical Imaging* 10, 4 (2023), 044503–044503.
- [26] Gregory Holste, Song Wang, Ajay Jaiswal, Yuzhe Yang, Mingquan Lin, Yifan Peng, and Atlas Wang. 2023. CXR-LT: Multi-Label Long-Tailed Classification on Chest X-Rays. (2023).
- [27] Gregory Holste, Song Wang, Ziyu Jiang, Thomas C Shen, George Shih, Ronald M Summers, Yifan Peng, and Zhangyang Wang. 2022. Long-tailed classification of thorax diseases on chest x-ray: A new benchmark study. In *MICCAI Workshop on Data Augmentation, Labelling, and Imperfections*. Springer, 22–32.
- [28] Yaxin Hu, Hajin Lim, Hailey L Johnson, Josephine M O'Shaughnessy, Lisa Kakonge, Lyn Turkstra, Melissa Duff, Catalina Toma, and Bilge Mutlu. 2023. Investigating Day-to-day Experiences with Conversational Agents by Users with Traumatic Brain Injury. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–15.
- [29] Shih-Hong Huang, Chieh-Yang Huang, Ya-Fang Lin, and Ting-Hao Kenneth Huang. 2023. What Types of Questions Require Conversation to Answer? A Case Study of AskReddit Questions. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–9.
- [30] Zhongzhen Huang, Xiaofan Zhang, and Shaoting Zhang. 2023. KiUT: Knowledge-injected U-Transformer for Radiology Report Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 19809–19818.
- [31] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *Comput. Surveys* 55, 12 (2023), 1–38.
- [32] Eunhyung Jo, Daniel A Epstein, Hyunhoon Jung, and Young-Ho Kim. 2023. Understanding the benefits and challenges of deploying conversational AI leveraging large language models for public health intervention. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [33] Alistair EW Johnson, Tom J Pollard, Nathaniel R Greenbaum, Matthew P Lungren, Chih-ying Deng, Yifan Peng, Zhiyong Lu, Roger G Mark, Seth J Berkowitz, and Steven Horng. 2019. MIMIC-CXR-JPG, a large publicly available database of labeled chest radiographs. *arXiv preprint arXiv:1901.07042* (2019).
- [34] Emma K Jones, Alyssa Banks, Genevieve B Melton, Carolyn M Porta, and Christopher J Tignanelli. 2022. Barriers to and facilitators for acceptance of comprehensive clinical decision support system-driven care maps for patients with thoracic trauma: interview study among health care providers and nurses. *JMIR human factors* 9, 1 (2022), e29019.
- [35] Kaggle. 2021. *VinBigData Chest X-ray Abnormalities Detection*. <https://www.kaggle.com/competitions/vinbigdata-chest-xray-abnormalities-detection>
- [36] Maxime Kayser, Cornelius Emde, Oana-Maria Camburu, Guy Parsons, Bartłomiej Papież, and Thomas Lukasiewicz. 2022. Explaining chest x-ray pathologies in natural language. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 701–713.
- [37] Sunnie SY Kim, Elizabeth Anne Watkins, Olga Russakovsky, Ruth Fong, and Andrés Monroy-Hernández. 2023. "Help Me Help the AI": Understanding How Explainability Can Support Human-AI Interaction. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [38] Ajay Kohli and Saurabh Jha. 2018. Why CAD failed in mammography. *Journal of the American College of Radiology* 15, 3 (2018), 535–537.
- [39] Himabindu Lakkaraju, Dylan Slack, Yuxin Chen, Chenhao Tan, and Sameer Singh. 2022. Rethinking Explainability as a Dialogue: A Practitioner's Perspective. *arXiv preprint arXiv:2202.01875* (2022).
- [40] Ricardo Bigolin Lanfredi, Mingyuan Zhang, William Auffermann, Jessica Chan, Phuong-Anh Duong, Vivek Srikumar, Trafton Drew, Joyce Schroeder, and Tolga Tasdizen. 2021. REFLAX: Reports and eye-tracking data for localization of abnormalities in chest x-rays.
- [41] Jason J Lau, Soumya Gayen, Asma Ben Abacha, and Dina Demner-Fushman. 2018. A dataset of clinically generated visual questions and answers about radiology images. *Scientific data* 5, 1 (2018), 1–10.
- [42] Yi-Chieh Lee, Yichao Cui, Jack Jamieson, Wayne Fu, and Naomi Yamashita. 2023. Exploring effects of chatbot-based social contact on reducing mental illness stigma. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [43] Christian Leibig, Moritz Brehmer, Stefan Bunk, Danalyn Byng, Katja Pinker, and Lale Umutlu. 2022. Combining the strengths of radiologists and AI for breast

- cancer screening: a retrospective analysis. *The Lancet Digital Health* 4, 7 (2022), e507–e519.
- [44] Chunyuan Li, Cliff Wong, Sheng Zhang, Naoto Usuyama, Haotian Liu, Jianwei Yang, Tristan Naumann, Hoifung Poon, and Jianfeng Gao. 2023. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. *arXiv preprint arXiv:2306.00890* (2023).
 - [45] Zhuoyang Li, Minhui Liang, Hai Trung Le, Ray Lc, and Yuhan Luo. 2023. Exploring Design Opportunities for Reflective Conversational Agents to Reduce Compulsive Smartphone Use. In *Proceedings of the 5th International Conference on Conversational User Interfaces*. 1–6.
 - [46] Lizi Liao, Le Hong Long, Zheng Zhang, Minlie Huang, and Tat-Seng Chua. 2021. MMConv: an environment for multimodal conversational search across multiple domains. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval*. 675–684.
 - [47] Q Vera Liao, Daniel Gruen, and Sarah Miller. 2020. Questioning the AI: informing design practices for explainable AI user experiences. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–15.
 - [48] Ruizhi Liao, Geeticka Chauhan, Polina Golland, S Berkowitz, and Steven Horng. 2021. Pulmonary edema severity grades based on MIMIC-CXR (version 1.0. 1). *PhysioNet* (2021).
 - [49] Fenglin Liu, Changchang Yin, Xian Wu, Shen Ge, Ping Zhang, and Xu Sun. 2021. Contrastive Attention for Automatic Chest X-ray Report Generation. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*. 269–280.
 - [50] Yunjie Liu, WF Goevaerts, Max V Birk, Harel Kempers, and Yuan Lu. 2022. Development of a Conversational Dietary Assessment Tool for Cardiovascular Patients. In *International Conference on Human-Centred Software Engineering*. Springer, 179–190.
 - [51] Wookjae Maeng and Joonhwan Lee. 2022. Designing and evaluating a chatbot for survivors of image-based sexual abuse. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–21.
 - [52] Elliot Mbunge and John Batani. 2023. Application of deep learning and machine learning models to improve healthcare in sub-Saharan Africa: Emerging opportunities, trends and implications. *Telematics and Informatics Reports* (2023), 100097.
 - [53] Carlo Metta, Andrea Beretta, Riccardo Guidotti, Yuan Yin, Patrick Gallinari, Salvatore Rinzivillo, and Fosca Giannotti. 2023. Improving trust and confidence in medical skin lesion diagnosis through explainable deep learning. *International Journal of Data Science and Analytics* (2023), 1–13.
 - [54] Tim Miller. 2019. Explanation in artificial intelligence: Insights from the social sciences. *Artificial intelligence* 267 (2019), 1–38.
 - [55] Ha Q Nguyen, Khanh Lam, Linh T Le, Hieu H Pham, Dat Q Tran, Dung B Nguyen, Dung D Le, Chi M Pham, Hang TT Tong, Diep H Dinh, et al. 2022. VinDr-CXR: An open dataset of chest X-rays with radiologist's annotations. *Scientific Data* 9, 1 (2022), 429.
 - [56] Liqiang Nie, Fangkai Jiao, Wenjie Wang, Yinglong Wang, and Qi Tian. 2021. Conversational image search. *IEEE Transactions on Image Processing* 30 (2021), 7732–7743.
 - [57] Alisha Pradhan and Amanda Lazar. 2021. Hey Google, do you have a personality? Designing personality and personas for conversational agents. In *Proceedings of the 3rd Conference on Conversational User Interfaces*. 1–4.
 - [58] Xiang-Zhi Qiu, Chien Wen Yuan, Nanyi Bi, Ming-Chyi Huang, and Chuang-Wen You. 2023. Exploring the Challenges and Opportunities in Developing Systems to Improve Alcohol Use Disorder through Chatbot Technology. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–5.
 - [59] Paisan Ruamviboonsuk, Jonathan Krause, Peranot Chotcomwongse, Rory Sayres, Rajiv Raman, Kasumi Widner, Bilson JL Campana, Sonia Phene, Kornwipa Hemarat, Mongkol Tadarati, et al. 2019. Deep learning versus human graders for classifying diabetic retinopathy severity in a nationwide screening program. *NPJ digital medicine* 2, 1 (2019), 25.
 - [60] Cassandra EL Seah, Zheyuan Zhang, Sijin Sun, Esther Wiskerke, Sarah Daniels, Talya Porat, and Rafael A Calvo. 2022. Designing Mindfulness Conversational Agents for People With Early-Stage Dementia and Their Caregivers: Thematic Analysis of Expert and User Perspectives. *JMIR aging* 5, 4 (2022), e40360.
 - [61] Korok Sengupta, Sayan Sarcar, Alisha Pradhan, Roisin McNaney, Sergio Sayago, Debaleena Chattopadhyay, and Anirudha Joshi. 2020. Challenges and opportunities of leveraging intelligent conversational assistant to improve the well-being of older adults. In *Extended abstracts of the 2020 CHI conference on human factors in computing systems*. 1–4.
 - [62] Poonam Shourie, Vatsala Anand, and Shefali Gupta. 2023. An Efficient CNN Framework for Radiologist level Pneumonia Detection Using Chest X-ray Images. In *2023 3rd International Conference on Intelligent Technologies (CONIT)*. IEEE, 1–6.
 - [63] Karan Singhal, Shekoofeh Azizi, Tao Tu, S Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, et al. 2023. Large language models encode clinical knowledge. *Nature* 620, 7972 (2023), 172–180.
 - [64] Dylan Slack, Satyapriya Krishna, Himabindu Lakkaraju, and Sameer Singh. 2023. Explaining machine learning models with interactive natural language conversations using TalkToModel. *Nature Machine Intelligence* (2023), 1–11.
 - [65] Jens B Stephansen, Alexander N Olesen, Mads Olsen, Aditya Ambati, Eileen B Leary, Hyatt E Moore, Oscar Carrillo, Ling Lin, Fang Han, Han Yan, et al. 2018. Neural network analysis of sleep stages enables efficient diagnosis of narcolepsy. *Nature communications* 9, 1 (2018), 5229.
 - [66] Leo K Tam, Xiaosong Wang, Evrim Turkbey, Kevin Lu, Yuhong Wen, and Daguang Xu. 2020. Weakly Supervised One-Stage Vision and Language Disease Detection Using Large Scale Pneumonia and Pneumothorax Studies. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part IV*. 45–55.
 - [67] Tim Tanida, Philip Müller, Georgios Kaissis, and Daniel Rueckert. 2023. Interactive and Explainable Region-guided Radiology Report Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7433–7442.
 - [68] Omkar Thawkar, Abdelrahman Shaker, Sahal Shaji Mullappilly, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Jorma Laaksonen, and Fahad Shabbaz Khan. 2023. Xraygpt: Chest radiographs summarization using medical vision-language models. *arXiv preprint arXiv:2306.07971* (2023).
 - [69] Viswanath Venkatesh, Michael G Morris, Gordon B Davis, and Fred D Davis. 2003. User acceptance of information technology: Toward a unified view. *MIS quarterly* (2003), 425–478.
 - [70] Himanshu Verma, Jakob Mlynar, Roger Schaer, Julien Reichenbach, Mario Jreige, John Prior, Florian Evéquo, and Adrien Depeursinge. 2023. Rethinking the role of AI with physicians in oncology: revealing perspectives from clinical and research workflows. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–19.
 - [71] Lu Wang, Chaomei Chen, and Jina Huh-Yoo. 2023. Investigating the Synonyms of Conversational Agents to Aid Cross-Disciplinary CA Research. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–10.
 - [72] Zhanyu Wang, Hongwei Han, Lei Wang, Xiu Li, and Luping Zhou. 2022. Automated radiographic report generation purely on transformer: A multicriteria supervised approach. *IEEE Transactions on Medical Imaging* 41, 10 (2022), 2803–2813.
 - [73] Zhanyu Wang, Lingqiao Liu, Lei Wang, and Luping Zhou. 2023. METransformer: Radiology Report Generation by Transformer with Multiple Learnable Expert Tokens. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 11558–11567.
 - [74] Julia Wolleb, Florentin Bieder, Robin Sandkühler, and Philippe C Cattin. 2022. Diffusion models for medical anomaly detection. In *International Conference on Medical image computing and computer-assisted intervention*. Springer, 35–45.
 - [75] Yaxiong Wu, Craig Macdonald, and Iadh Ounis. 2022. Multi-Modal Dialog State Tracking for Interactive Fashion Recommendation. In *Proceedings of the 16th ACM Conference on Recommender Systems*. 124–133.
 - [76] Yao Xie, Melody Chen, David Kao, Ge Gao, and Xiang'Anthony' Chen. 2020. CheXplain: enabling physicians to explore and understand data-driven, AI-enabled medical imaging analysis. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
 - [77] Honglin Xiong, Sheng Wang, Yitao Zhu, Zihao Zhao, Yuxiao Liu, Qian Wang, and Dinggang Shen. 2023. Doctorglm: Fine-tuning your chinese doctor is not a herculean task. *arXiv preprint arXiv:2304.01097* (2023).
 - [78] Shawn Xu, Lin Yang, Christopher Kelly, Marcin Sieniek, Timo Kohlberger, Martin Ma, Wei-Hung Weng, Attila Kiraly, Sahar Kazemzadeh, Zakkai Melamed, et al. 2023. ELIXR: Towards a general purpose X-ray artificial intelligence system through alignment of large language models and radiology vision encoders. *arXiv preprint arXiv:2308.01317* (2023).
 - [79] Qian Yang, Yuexing Hao, Kexin Quan, Stephen Yang, Yiran Zhao, Volodymyr Kuleshov, and Fei Wang. 2023. Harnessing biomedical literature to calibrate clinicians' trust in AI decision support systems. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–14.
 - [80] Chenchen Ye, Lizi Liao, Suyu Liu, and Tat-Seng Chua. 2022. Reflecting on experiences for response generation. In *Proceedings of the 30th ACM International Conference on Multimedia*. 5265–5273.
 - [81] Li Yunxiang, Li Zihan, Zhang Kai, Dan Ruilong, and Zhang You. 2023. Chat-doctor: A medical chat model fine-tuned on llama model using medical domain knowledge. *arXiv preprint arXiv:2303.14070* (2023).
 - [82] Xiao Zhan, Yifan Xu, and Stefan Sarkadi. 2023. Deceptive AI Ecosystems: The Case of ChatGPT. In *Conversational User Interfaces, CUI'23, July 19–21, 2023, Eindhoven, Netherlands*.
 - [83] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. 2019. BERTScore: Evaluating Text Generation with BERT. In *International Conference on Learning Representations*.
 - [84] Xiaoman Zhang, Chaoyi Wu, Ya Zhang, Weidi Xie, and Yanfeng Wang. 2023. Knowledge-enhanced visual-language pre-training on chest radiology images. *Nature Communications* 14, 1 (2023), 4542.

- [85] Yuhao Zhang, Daisy Yi Ding, Tianpei Qian, Christopher D Manning, and Curtis P Langlotz. 2018. Learning to Summarize Radiology Findings. In *Proceedings of the Ninth International Workshop on Health Text Mining and Information Analysis*. 204–213.
- [86] Yuhao Zhang, Derek Merck, Emily Tsai, Christopher D Manning, and Curtis Langlotz. 2020. Optimizing the Factual Correctness of a Summary: A Study of Summarizing Radiology Reports. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. 5108–5120.
- [87] Zihao Zhao, Sheng Wang, Jinchun Gu, Yitao Zhu, Lanzhuju Mei, Zixu Zhuang, Zhiming Cui, Qian Wang, and Dinggang Shen. 2023. ChatCAD+: Towards a Universal and Reliable Interactive CAD using LLMs. *arXiv preprint arXiv:2305.15964* (2023).
- [88] Jiawei Zhou, Yixuan Zhang, Qianni Luo, Andrea G Parker, and Munmun De Choudhury. 2023. Synthetic lies: Understanding ai-generated misinformation and evaluating algorithmic and human solutions. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–20.